Forest Policy and Economics 143 (2022) 102817



Contents lists available at ScienceDirect

Forest Policy and Economics

journal homepage: www.elsevier.com/locate/forpol





Potential impacts of the proposed EU regulation on deforestation-free supply chains on smallholders, indigenous peoples, and local communities in producer countries outside the EU

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ARTICLE INFO

Keywords
Deforestation-free products
EU regulation
Smallholders
Indigenous people
Local communities
Producing countries

ABSTRACT

According to the new European Commission proposal for a regulation on deforestation-free products, six commodities - cattle, wood, palm oil, soy, cocoa, and coffee and their derivate products – grown or raised on land that was subject to deforestation or forest degradation will be banned from entering the EU market. In this commentary paper, we discuss the possible unintended effects of the regulation and elaborate on potential ways to reduce the negative impacts, on vulnerable groups in producer countries, such as smallholders, Indigenous peoples, and local communities (IPLCs). Such impacts could comprise exclusion of smallholders and IPLCs from high-value supply chains of commodities covered by the regulation, inadequate price premiums to cover the costs of complying with this regulation, increase the risk of land conflicts between large-scale and small-scale agriculture, push smallholders and IPLCs to marginal lands, and even exacerbate human rights risks already associated with large-scale commodity production. In order to avoid or minimize such impacts, we propose: 1) to make provisions for companies, smallholders, and producing regions to receive sufficient support in their transition to deforestation-free supply chains and to provide incentives so that this transition can be inclusive; 2) to further promote and strengthen holistic approaches, such as jurisdictional or other integrated landscape management approaches to address the many issues that can arise from conflicting interests and unequal power distribution in affected landscapes; 3) to make clear provisions for protection of human rights and to promote land rights for smallholders and IPLCs related to tenure security; and 4) to ensure that regular and robust impact assessments inform policy about both direct and indirect impacts of the regulation on smallholders and IPLCs so that timely and adequate action can be taken to minimize the unintended effects of this regulation. Addressing potential negative impacts on smallholders and IPLCs would also ensure that the progress achieved so far in terms of other UN SDGs related to poverty, inequality, hunger, and land restoration is not compromised.

1. Introduction

1.1. The current EU regulation proposal and what is missing in it

The new European Commission proposal for a regulation on deforestation-free products, announced on November 17, 2021, includes a set of measures with the purpose of minimizing the contribution of the EU to deforestation and forest degradation worldwide and reducing EU's contribution to greenhouse gas emissions and global biodiversity loss (European Commission, 2021c). According to the proposed regulation, six commodities - cattle, wood, palm oil, soy, cocoa, and coffee and their derivate products – grown or raised on land that was subject to deforestation or forest degradation after December 31, 2020, the planned "cut-off date" of the regulation, will be banned from entering the EU

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https://doi.org/10.1016/j.forpol.2022.102817
Received 1 April 2022; Received in revised form 11 July 2022; Accepted 7 August 2022
Available online 13 August 2022
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market after the regulation enters into force. In addition, the six commodities included in the regulation (referred to as "relevant commodities" in the text of the proposal) will have to comply with the producer countries' respective legislations on deforestation (European Commission, 2021c). The regulation proposal was welcomed by many as a groundbreaking measure to address global deforestation and forest degradation trends (ClientEarth, 2021a; FERN, 2021; Marzano, 2021).

Globally, total greenhouse gas emissions due to agriculture, forestry, and other land use represented 23% of total net anthropogenic emissions during 2007–2016 (IPCC, 2019), whereby measures to reduce forest degradation and deforestation are an important part of the fight against climate change. In 2017, 1.3 million ha of tropical deforestation was associated with international trade, of which the EU was responsible for 16%, making it the second largest importer of tropical deforestation after China (24%) (WWF, 2021). Overall, 5 million ha of tropical forests per year between 2005 and 2007 were lost due to expansion of large-scale and small-scale agriculture (WWF, 2021). Commodity production is a major driver of deforestation in many parts of the world: in Latin America, pasture was the major post-deforestation land use form (72%) followed by large-scale crop production (11%), while in Africa, small-scale croplands (61%) and pasture (15%) were the most important follow-up land uses, and in Asia, small-scale crop production (35%) and tree crops (28%) occupied deforested areas (De Sy et al., 2019). Since regulations and their enforcement in producer countries are not sufficient to halt deforestation and because a large part of this deforestation is linked to international trade, the demand-side measures, such as the currently proposed EU regulation, are believed to play a very important role in the future fights against deforestation (WWF, 2021). On the other hand, although beyond the scope of this commentary, past experience with ecologically friendly or sustainable production schemes does not necessarily show that these measures were successful in reducing deforestation (Alix-Garcia and Gibbs, 2017; van der Ven et al., 2018). In their comparative study of cases in Brazil, Indonesia, and Côte d'Ivoire, van der Ven et al. (2018) found that eco-labels and certification schemes for agricultural commodities were not effective in stopping conversion of land from forest to agriculture, which the authors attributed to a lack of wider uptake and the existence of regulatory loopholes in those schemes (van der Ven et al., 2018). Also, past evidence on the effectiveness of zero-deforestation in terms of avoided deforestation has not been well studied for key production regions of forest-risk commodities, making it difficult to establish a clear causal link between such measures and deforestation (Ingram et al., 2020).

As with any policy measure however, especially of this importance, there are likely to be unintended consequences, which are just as important as the intended ones. In tropical countries, smallholders, often including Indigenous peoples and local communities (IPLCs), play a crucial role in addressing climate change and forest conservation (Fair Trade Advocacy Office et al., 2021). Approximately one-quarter of global forests, or 1 billion hectares, are estimated to be managed by IPLCs (Garnett et al., 2018). Overall, there are about 500 million farms worldwide with a farm area of less than 2 ha, 40% of which sell a considerable amount of their produce in the markets (Lowder et al., 2016). The livelihoods of such smallholders are highly vulnerable in many cases (Grabs et al., 2021) and are often dependent on their integration into global supply chains, with better integration being linked to increased capability of smallholders to achieve desired livelihood and socio-ecological outcomes (Dou et al., 2020).

Public policies that recognize the complexity of the challenges faced by small farms are key to achieve SDG targets for ending hunger and reducing inequalities (Lowder et al., 2014). If the proposed regulation has negative consequences on the livelihoods of smallholders, Indigenous peoples, and local communities in the producer countries, then it would diminish the progress achieved so far in relation to Sustainable Development Goals, e.g. SDG 1 (no poverty), SDG 2 (Zero hunger) with Target 2.3 in particular, and SDG 10 (reduced inequalities). Unless the global approaches for food system transformation aiming at environmental and climate change outcomes put livelihoods at the center of such approaches, the risk of reaching these objectives "on the back of the rural poor" remains high (Davis et al., 2022).

At the same time, the current proposal of the regulation (European Commission, 2021c) does not specify any measures to support smallholders to meet the established requirements. It also does not require the operators to invest efforts in including smallholders and providing them with support to comply with due diligence requirements. Even though the disadvantages for smallholders are mentioned in both the impact assessment report (European Commission, 2021a) and in the introductory text of the regulation (European Commission, 2021c), the only proposed measure seems to be the planned cut-off date. According to the impact assessment report (European Commission, 2021a), part 1, p.62: "Whilst the long term impacts on third countries are expected to be positive, initial short term impacts caused by EU operators shortening/simplifying supply chains, reducing their number of suppliers and/or switching to lower-risk supply chains may particularly impact smallholders… Again, the suggested cut-off date of 2020 would significantly mitigate potentially negative social impacts by limiting the number of smallholders that would be caught working on land whose products cannot be sold to the EU — and ensuring that nearly all current commodity production from exporting countries can still make the cut".

In this commentary paper we discuss the possible unintended effects of the regulation on vulnerable groups in producer countries, such as smallholders, Indigenous peoples, and local communities as well as potential ways to reduce the negative impacts.

2. The extent of smallholder involvement in forest-risk commodity supply chains

Previous literature contains various definitions of smallholder and family farms based on the land size that they operate, which varies widely depending on the region and context (Lowder et al., 2016). In Table 1, we summarized information related to smallholders' involvement in the production of coffee, cocoa, and palm oil in non-EU producer countries, which are the top exporters of these commodities to the EU. Due to variation of land sizes that are attributed to smallholder farms in different studies, we report farm sizes for each country that a cited literature source on smallholder involvement in commodity production in Table 1 refers to. We focus on these three commodities because smallholder participation in their production is likely to be higher than in the production of the rest of the commodities included in the regulation proposal.

Together, nine countries supplied 85% of all coffee that was imported by the European Union in 2020 (Table 1). Coffee production in these countries is concentrated in the hands of smallholders, starting from 38% of output produced by smallholders in Brazil and reaching 70–95% of output in Honduras, Ethiopia, Uganda, and Vietnam. The case of cocoa is similar to coffee. Around 90% of cocoa production relies on 5–6 million smallholders in developing countries (European Commission, 2021a). The six top exporter countries of cocoa to the EU covered about 80% of cocoa imported by the EU in 2020. Up to two million smallholders in West Africa and 1.4 million small farms in Indonesia were involved in the production of cocoa (Table 1). Even in the palm oil sector, the involvement of smallholders is significant. The six top exporter countries of palm oil supplied 97% of total EU imports of the commodity in 2020, where 73% is covered just by two countries, Indonesia and Malaysia. Considerable share of oil palm plantations (28 to 42% in Malaysia and Indonesia respectively) in these two countries is still operated by smallholder farms.

Although production of beef, soya, and wood globally also involves smallholders and family farms, their relative importance when it comes to the quantities exported to the EU are expected to be much smaller compared to the cases of cocoa, coffee, and palm oil. For instance, in Brazil, 90% of soy is produced by large agricultural holdings (European Commission, 2021a).

Table 1
Major exporters of coffee, cocoa, and palm oil to EU, and share of smallholders in commodity production in 2020.

Commodity	Top exporters to the EU	Import quantity, (tonnes)	% of extra-EU imports	Smallholder involvement in total commodity production
Coffee	Brazil	933,000	32%	In total, 75% of coffee growers or 300,000 smallholder farms operate on less than 5 ha of land and supply 38% of total output (IDH, 2021).
	Vietnam	642,000	22%	With farm size less than 1 ha, 600,000 smallholders produce 95% of coffee (IDH, 2021).
	Honduras	216,000	7%	M70% of output) is produced by farms smaller than 2 ha (70% of farmers) and between 2 and 7 ha (25% of farmers) (IDH, 2021).
	Uganda	166,000	6%	About 1.7 million smallholders farms of size between 0.5 and 3 ha produce 85% of coffee output (IDH, 2021).
	Colombia	154,000	5%	With farm sizes between 2 and 5 ha, 560,000 smallholders produce 69% of coffee; while 95% of coffee growers are smallholders (IDH, 2021).
	India	108,000	4%	Coffee growers include 377,000 land holdings of less than 10 ha and 2900 of more than 10 ha (Coffee Board of India, 2020).
	Peru	96,000	3%	About 60% of coffee farms are smallholders (132000) with farm sizes of around 2 ha (IDH, 2021).
	Ethiopia	78,000	3%	With an average farm size between 0.5 and 2 ha, 2 million smallholders produce 90% of coffee (IDH, 2021).
	Indonesia	78,000	3%	1.5 million smallholders with less than 1 ha land manage 95% of coffee plantations (IDH, 2021).
Cocoa	Côte d'Ivoire	1,135,258	45%	About 70% of total globally produced cocoa beans comes from four countries in West Africa: Côte d'Ivoire, Ghana, Cameroon and Nigeria (Wessel and Quist-Wessel, 2015), produced by about 1.8 to 2 million smallholders (Schulte et al., 2020).
	Ghana	369,164	15%	
	Cameroon	206,044	8%	
	Nigeria	151,073	6%	
	Ecuador	68,960	3%	No concrete data on the number of smallholders involved in cocoa production in Ecuador were available
Palm oil	Indonesia	43,220	2%	Over 1.4 million smallholder farmers contribute 93% of national production (VECO, 2011).
	Indonesia	3,381,079	48%	In Indonesia, smallholders control 42% of national palm oil planted land, and in Malaysia, smallholders manage 28% of palm oil plantations (European Commission, 2021a).
	Malaysia	1,793,909	25%	
	Papua New Guinea	451,120	6%	In 2010, 23,000 small farmers with an average size of 6 ha operated 40% of total oil plantation area (Elahi and Michael, 2017).
	Guatemala	430,727	6%	Both production and processing of palm oil in Guatemala are controlled by six large national business groups (Guereña and Zepeda, 2013).
	Colombia	429,653	6%	Around 70% of all oil palm properties had the area size of less than 20 ha, but together controlled only 8.2% of total oil palm plantation area in 2011 (Potter, 2020).
	Honduras	392,781	6%	Approximately 17,000 small-scale producers grow oil palm on areas less than 10 ha land each (Proforest, 2016).

Source: Eurostat for import quantities and shares.

3. Potential unintended impacts of the regulation on smallholders and IPLCs

Challenges to successful engagement of smallholders to implement "zero deforestation" commitments by corporations in the past included supply chain complexity, weak local governance and institutions, heterogeneity of smallholders, and smallholder vulnerability in terms of short-term concerns (Bakhtary et al., 2020). On the other hand, smallholders exhibited a limited capacity to comply with measures promoting sustainable supply chains in the past. The reasons for this limited compliance include low education levels, limited financial resources, land tenure issues such as unclear rights and tenure insecurity, limited access to information, low degree of organization, and risk-averse attitudes of smallholders towards sustainable land use systems that require long-term commitment (Ansah et al., 2020; Brandi et al., 2015; Grabs et al., 2021; Hoang et al., 2019; Nguyen et al., 2018). Given these concerns, several ways in which smallholders can be affected by this regulation are explained below.

3.1. Exclusion of smallholders from the international supply chains of relevant commodities

During the targeted stakeholder consultations conducted as part of

the impact assessment (European Commission, 2021a), some participants had concerns that due diligence legislation could disengage smallholders because of the associated burden in terms of high transaction costs, which could in turn lead to additional deforestation from loss of livelihood. Loss of livelihood could occur if smallholders are excluded from value chains of relevant commodities. Given limited alternative income opportunities, there is a risk of further deforestation by smallholders for subsistence purposes. Analysing land use and land cover changes in coffee-growing areas over 2000–2010 in Central Vietnam, Meyfroidt et al. (2013) show that deforestation was caused by shifting cultivation by ethnic minorities and poor migrants, but that displacement of shifting cultivation into forest edges took place in response to expansion of coffee and other commercial crops in agricultural areas (Meyfroidt et al., 2013).

The case studies also confirm that the smallholders face the risk of being excluded from the value chain due to the complexity in implementing traceability systems, because the EU operators are likely to react to the new regulation by shortening and simplifying supply chains (European Commission, 2021a). Shortening or simplifying supply chains could be done by reducing the number of smallholders and instead sourcing commodities from larger agricultural farms or by expansion of own farms operated by exporting companies. This could reduce transaction costs for companies, but reduce the number of smallholders participating in high value supply chains.

The smallholders engaged in cocoa, coffee, and palm oil production are expected to be affected to a significant degree due to their overall contribution to global supply of these commodities. The examples of smallholders being excluded from high value supply chains due to higher export standards have been documented in many papers. For instance, Okello et al. (2011) report that smallholders' contribution to green bean production in Kenya reduced from 60% to 30% during the period of 1980 to 2003 as the leading exporters dropped more than half of the smallholder outgrowers in order to meet the demands of strict compliance with the private food safety protocols of EU retailers to whom they supplied. The same outcome was observed in the Senegalese bean sector, with smallholder procurement being reduced from 95% in 1999 to 52% in 2005 as the largest companies started their own estate farms in order to comply with EurepGAP certification requirements (Maertens et al., 2009). Depending on the extent to which smallholders' output of relevant commodities was exported to the EU prior to the proposed regulation, the exclusion of smallholders from these high-value supply chains will lead to reduced rural income and increased inequality in some cases.

Experiences from other parts of the world show that excessive regulations resulting in very high transaction costs can make adherence to legal requirements impractical for many forest users, especially for community-based and small and medium size enterprises (ITTO and FAO, 2020). For instance, the costly management plans required by the 2003 General Law for Sustainable Forest Development in Mexico led owners of secondary forests to allocate their land to agricultural activities rather than to forest management or restoration (Román-Dañobeytia et al., 2014). Small-scale forest users and community-based organizations were often poorly prepared to comply with complex administrative requirements related to sustainable use of forests (ITTO and FAO, 2020; Román-Dañobeytia et al., 2014). Moreover, the area of forest under long-term management plans in many countries remains low compared to Europe and North and Central America: in Africa, less than 25% of forests and in South America, less than 20% of forest had management plans (FAO, 2020a). Majority of certified forest areas are also found in Europe and North America, while the forest area under certification remains low in other parts of the world (FAO, 2020b). The latter indicates that price premiums associated with certified forest products may not have had any significant impact in producing landscapes given their limited coverage of total forest area. High transaction costs associated with certification are also often reported for other commodities included in the EU regulation proposal. For instance, in the oil palm sector, independent smallholders participating in the Roundtable on Sustainable Palm Oil (RSPO) certification program reported very high costs related to certification, such as establishment of an internal control system (ICS), acquiring land titles, preparation of documents for individual farmers and for farmer groups, as well as audit preparations (Apriani et al., 2020; Brandi et al., 2015). Without external support, smallholders would be unable to cover all the costs associated with RSPO certification (Apriani et al., 2020). A further question that can be raised is whether EU demand for relevant commodities will decrease in response to higher prices for these commodities leading to less amount exported by producer countries and eventually less income earned from the sale of relevant commodities.

3.2. *Price premium: Who will receive it and will it be enough?*

According to the impact assessment report, operators in third countries, including smallholders, could face costs to develop or implement systems to comply with the new requirements, whereby these costs could be passed through the prices of products (European Commission, 2021a). One assumption is that the increase in product prices could serve as a "price premium" for complying with deforestation-free requirements. However, an increase in the prices of products does not necessarily mean that the prices obtained by smallholders will increase to the same extent. For instance, in the case of coffee growers in Nicaragua, neither Fairtrade nor organic certification had a significant impact on farmers' income (Jena et al., 2017). Neilson (2008) argues that the increased costs of new arrangements coming from the global private regulation of ethical and environmental standards for coffee production and trade may have actually exerted a downward pressure on farm-gate prices in Indonesia. In the case of timber producers in Vietnam, the costs of FSC certification were still too expensive for most smallholders, especially the audit fee paid to the certification agency (Hoang et al., 2019; Nguyen et al., 2018). In a review of smallholder forestry in the FSC system, Bulkan (2020) reports that market benefits from FSC for smallholders in the global South have been very small and insufficient to outweigh the costs of certification, which is why many smallholders considered retaining FSC certification only if there was donor support and/or if markets were secure (Bulkan, 2020).

In general, farmers often receive very little of the final product value generated from forest-risk commodities. The smallest share among the six relevant commodities is observed for cocoa farmers who receive only 3 to 7% of the retail price of a chocolate bar compared to 40% and 35% of the price captured by chocolate manufacturers and retailers respectively (Brack, 2019). Different analyses of cash crop value chains show that farmers receive often the smallest share of the profit margins generated along the value chains. For instance coffee growers in Côte d'Ivoire received 13.8% of the retail price, while growers in Costa Rica received 14.6% in 1994 (Ponte, 2002). There is hardly any recent evidence on the distribution of retail prices across the value chain of relevant commodities.

Moreover, high costs of complying with export standards can lead to exporter consolidation with a risk of generating a monopsonistic environment, where only one buyer (e.g., an exporter) faces many suppliers (e.g., small farmers) (Neilson, 2008). Such is the case of smallholder producers of palm oil in the Ixcán municipality in Guatemela who depend on one single company to buy and process their palm fruit with no price guarantees (Guereña and Zepeda, 2013). Similar monopsony structures in palm oil sector were also reported for Papua New Guinea (Anderson, 2006; Elahi and Michael, 2017).

As mentioned earlier, different hurdles faced by smallholders have limited their participation in certification schemes related to sustainable supply chains in the past (Ansah et al., 2020; Brandi et al., 2015; Grabs et al., 2021; Hoang et al., 2019; Nguyen et al., 2018), even before the new EU regulation on deforestation-free products was proposed. Given the limited options available for smallholders and IPLCs, a potential outcome of being excluded from EU-value chains would be selling to

non-EU markets with lax environmental restrictions. As the supply of commodities for export to non-EU countries increases, smallholders may have to settle for a lower price than previously, in addition to losing out on a price premium.

3.3. Possible land conflicts between small-scale and large-scale agriculture

As mentioned earlier, the supply chains of operators are expected to be shortened and simplified in response to the new regulation. In addition to excluding smallholders who are not able to comply, this generates incentives for large-scale agriculture, which does not only reinforce existing inequalities between large-scale and small-scale producers (Gómez Tovar et al., 2005), but could also potentially lead to land conflicts between large scale agriculture and local communities. When weak land tenure rights are combined with governments favoring land concessions to large-scale agriculture, land contestations can occur which often work against Indigenous peoples and local communities, in many cases with severe negative impacts on livelihoods and cultural losses, such as in cases in East Africa, Indonesia and the Mekong (Pacheco et al., 2018, 2021). For instance, in Vietnam, reports indicate the use of coercive means to get smallholders contribute their lands to large-scale rubber enterprises in return for shareholding, with other cases also involving land appropriations from smallholders in favor of commercial investors in cash crops such as coffee or rubber (Hirsch et al., 2015). In Guatemala, reports have been made about indirect pressure applied by large scale oil palm companies to get smallholders to sell their land below market prices (Guereña and Zepeda, 2013). Soy production has also been linked to land grabbing in several countries in South America (WWF, 2014).

If the current land conflicts between the large-scale and small-scale agriculture are exacerbated due to the proposed EU regulation, these in turn can push the smallholders and IPLCs to forest frontiers resulting in deforestation or to marginal lands with worsened livelihoods.

3.4. Push of smallholders and local communities towards marginal lands and their reduced economic independence

After the regulation enters into force, commodities produced on deforested land or land that was subjected to forest degradation after December 31, 2020, will be banned from entering the EU markets. In the producer countries that supply the relevant commodities to the EU, this is likely to increase the demand by large-scale enterprises who have the capacity to comply for "clean" land that was already converted to agricultural use prior to the cut-off date. One potential effect from this is that land under other crops will be converted to the production of relevant commodities, while the production of commodities not included in the proposal, e.g. those for subsistence agriculture, might shift to other areas linked to further deforestation and forest degradation. In the past, expansion of forest-risk commodities have already been linked to displacement of food crops exacerbating the risk of food insecurity (Guereña and Zepeda, 2013), with reports of subsistence farming being pushed towards marginal lands in forested areas (SNV, n. d.). If such shifts take place, and the regulation indirectly exacerbates deforestation for subsistence agriculture, then the net effect of the regulation on deforestation will be probably much smaller than expected considering that subsistence agriculture is the second most important driver of deforestation after commercial agriculture (Hosonuma et al., 2012; Pacheco et al., 2021).

Even in the (unlikely) case that no further deforestation and forest degradation occurs, an increased demand for land to produce relevant commodities could shift focus away from diverse farming systems, such as agroforestry and home gardens which are important components of global forest landscape restoration initiatives (e.g. Bonn Challenge) that contribute to food security.

Finally, in areas where livelihood opportunities are already limited, losing access to land due to exacerbated land conflicts will force smallholders and IPLCs to seek wage labor in those large-scale farms responsible for their land loss or push them to migrate to other places in search of work. This can result in the loss of overall economic independence of smallholders and IPLCs.

3.5. Human rights risks

Land is not only a critical asset for livelihoods of rural populations, it is also a crucial basis for ensuring several human rights including the right to life, the right to food, housing, and property, and the right to development and self-determination (Tura, 2018; United Nations, 1948). Yet in many countries where the forest-risk commodities are produced, compliance with national laws does not necessarily provide protection for the rights of Indigenous peoples, forest defenders, and local communities (ClientEarth, 2021a). In contrary, examples from Indonesia, Brazil, and Ethiopia show that changes in national laws can legalize actions that exacerbate human rights risks and institutionalize dispossession of land from smallholders (ClientEarth, 2021a, 2021b; Jong, 2020; Tura, 2018). In some parts of Guatemala, oil palm plantation expansion has been linked to practices that violated basic human rights to free, prior, and informed consent, where indirect coercion into sale of land turned smallholders into temporary laborers (Guereña and Zepeda, 2013). Indigenous rights abuses related to large-scale palm oil production have been reported also in other countries that are major producers (Elahi and Michael, 2017). The proposed EU regulation does not require operators to follow international standards on tenure rights and Free, Prior and Informed Consent (FPIC). Without protecting the rights of local communities to manage and restore forested areas, the proposed regulation could reinforce unintendedly unequal access to and control over land due to pre-existing land tenure insecurity issues and could aggravate the human rights risks.

4. What can be done to reduce the potential negative impacts of the regulation on smallholders and IPLCs

Considering the potential unintended effects of the proposed regulation on smallholders in producer countries as outlined above, the following steps are proposed to minimize the impact of this regulation on other sustainable development goals.

4.1. Incentives for operators, capacity-building for smallholders, and supporting "readiness" of producer regions

A policy measure promoting sustainable value chains such as the proposed regulation should be more specific about incentives for suppliers and smallholders to upgrade their production practices across supply chains (Pacheco et al., 2018). The current regulation proposal does not specify any incentives or type of support for smallholders, companies, or producing regions. Capacity building in terms of sustainable farm management practices is required for smallholders involved in forest risk commodities who generally have lower education and limited access to financial resources (Duguma et al., 2021; Marzano, 2021; Nguyen et al., 2018). Supporting farmers to form unions or groups could help reduce their transaction costs related to participation in sustainable supply chains, while improving access to credit can boost their capacity to undertake structural changes in their farming systems and to increase efficiency (Duguma et al., 2021). Improving farmers' access to credit is not an easy task considering the numerous challenges they face when applying for a formal credit, such as lack of collateral in the case of land tenure insecurity (Herliana et al., 2018), as well as risk aversion, high transaction costs, and financial illiteracy among borrowers (Balana et al., 2020). Therefore policies that enhance smallholders and IPLC's capacity to own a collateral, such as a land title or assets, can positively affect their access to credit (Balana and Oyeyemi, 2022). Moreover, community-led organizations could work together

with the credit institutions on improving farmers' risk perceptions with regards to formal credit. Finally, in cases where no collateral is available, group lending micro credit programs – where small size loans are provided to a group of individuals who do not have a bankable collateral (Schurmann and Johnston, 2009) – can be a viable solution to ensure that farmers have access to financial resources.

Forming farmer groups or associations can increase economies of scale which can benefit forest farmers and companies alike in terms of complying with zero deforestation requirements (Nguyen et al., 2018). Furthermore, any measures to improve farmers' technical efficiency could help reduce the demand for further agricultural land expansion. However, given the limited capacity of extension services, new cost-effective measures are required to help smallholders build capacity on more sustainable management practices. For instance, training-of-trainers and farmer-to-farmer extension have proven to be very effective in disseminating new technologies among farmers and are helpful where the capacity of extension services is limited (Akinnifesi et al., 2008).

In order to protect smallholders from being excluded from deforestation-free supply chains, additional incentives could be provided for operators to engage smallholders and to ensure a more inclusive transition towards sustainable supply chains. Although several company-driven smallholder engagement programs exist in cocoa and palm oil sectors, their impacts on smallholders are not clear, especially because companies use variety of approaches and often fail to define clear scale and scope for such programs (Bakhtary et al., 2020). Support for companies could include provision of specific guidance on how to set-up smallholder programs and assistance with improving traceability of supply chains e.g. through smallholder mapping.

4.2. Jurisdictional and landscape approaches

In addition to smallholders and companies, supporting producer regions or countries in order to improve their benchmarking score could help ensure a more inclusive transition to sustainable supply chains that the current regulation aims to achieve. Supporting the whole region in improving their sustainability practices also helps to minimize leakage of deforestation into other commodities or areas covered by crops not covered by this regulation. Jurisdictional or landscape approaches can ensure that the transformation into sustainable supply chains can take place in a holistic process involving all relevant actors in a region or a landscape, although these processes would require a long-term engagement in terms of financial support to build up trust and networks within and outside of jurisdictions and landscapes (Bakhtary et al., 2020; Seymour et al., 2020). Future commitments related to deforestation-free production, including the proposed regulation by the EU, will affect different industries that play a key role in many economies around the world, which is why preparing stakeholders for this transition and helping countries to take the necessary steps to meet the demand for deforestation-free commodities will be crucial (Pham et al., 2020).

Jurisdictional approaches (JA) refer to government-led multi-stakeholder processes that can bring different actors within certain administrative units, e.g., provinces and districts, to deliver a common goal, such as reduced or zero-deforestation, in conjunction with social and economic objectives while linking them to external financial and market incentives (Seymour et al., 2020). Financial support coming from external climate funding sources such as REDD+ program are critical for implementation of jurisdictional approaches (Garcia et al., 2021). The significance of jurisdictional approaches to support inclusion of smallholders in deforestation-free supply chains is already recognized by companies, where measures such as facilitating land registration and smallholder mapping have helped to reduce supply chain complexity in the past (Bakhtary et al., 2020). However, several constraints to successful JAs have been identified, including limited financial incentives to drive land-use change, lack of spatial data to guide better land-use planning, and limited capacity of local jurisdictions (Seymour et al., 2020). In addition, aligning JAs with national and local initiatives with potentially different views on resource use can become problematic, while the success of JAs depends highly on political willingness and the support of government at the local level, as well as on the availability of substantial financial resources (Essen and Lambin, 2021).

While a jurisdictional approach is an integrated landscape approach implemented within political boundaries of an administrative unit (Buchanan et al., 2019), different definitions of a "landscape", which can refer to ecological or spatial attributes to define conservation and development goals, make it difficult to agree on a common term for a "landscape approach"(Reed et al., 2015). To understand what a "landscape approach" aiming to integrate agricultural and environmental targets means, it is helpful to look at ten guiding principles proposed by Sayer et al. (2013): 1) continual learning and adaptive management; 2) common concern entry point; 3) addressing issues at multiple scales; 4) acknowledging multifunctionality of landscapes; 5) engaging multiple stakeholders; 6) negotiated and transparent change logic; 7) clarification of rights and responsibilities; 8) participatory and user-friendly monitoring; 9) maintaining and strengthening landscape resilience; and 10) strengthened stakeholder capacity.

Following these ten principles, landscape approaches would enable bringing all relevant stakeholders together to negotiate joint terms for transition into sustainable supply chains. However, all integrated landscape approaches, including jurisdictional approaches, will require long-term commitment to build trust and networks from both governments and external funding agencies (Seymour et al., 2020).

4.3. Protect human rights, promote tenure security

The proposed regulation needs to ensure that the national laws and international standards on tenure rights and Free, Prior and Informed Consent (FPIC) are considered within the benchmarking system to protect the IPLCs against land rights violations (FERN, 2021). Prioritizing local communities by protecting their rights to manage and restore forests can help bridge goals related to climate change mitigation, conservation, and sustainable development (Erbaugh et al., 2020). This could be done, for example, by promoting clear demarcation processes for communities and by facilitation to obtain land use certificates for smallholders. Increasing land tenure security can help smallholders to participate in sustainable supply chains and collaborations with processing companies (Nguyen et al., 2018). Clarifying land use, tenure, and access has been identified as one of the critical elements of any forest-based climate-change mitigation and adaptation policy measures (ITTO and FAO, 2020). While tenure reforms in recent past have generated some benefits for smallholders and communities, tenure reforms alone are not sufficient to secure benefits for local people from the commercial use of forest resources as these are often dependent on community capacity and market conditions (Pacheco, 2012). Furthermore, there are well-documented risks related to provision of land certificates, including high time and financial costs associated with the procedures, formalization of existing unequal distribution of land and opportunistic land acquisition by the elite (Benjaminsen et al., 2009), which the policy-makers need to be aware of.

4.4. Systematic and science-based impact assessments

In order to ensure the socio-economic compatibility of the regulation, the "general review of impacts on smallholders and IPLCs" provided for in Article 32 of the proposed regulation is very important. However, the wording of Article 32 does not include quality requirements for such reviews, which should go beyond qualitative reviews and include robust impact studies based on representativeness and adequate modeling of the cause-effect relationships between the intervention and the observed changes in producer countries in terms of both environmental and socio-economic indicators. The review of impacts on

smallholders and IPLCs should include not only direct impacts (such as to which extent participation of smallholders in deforestation-free supply chains was affected by the regulation, whether participation in deforestation-free supply chains affected their farm incomes and land use), but also indirect impacts. The indirect impacts could include changes in overall land use by smallholders and other actors in producing areas for commodities included in the regulation including whether shifts of land between commercial and subsistence agriculture took place, whether other land uses such as agroforestry systems or restoration practices were affected by the increased demand for "land free of deforestation and degradation", and whether any human rights violations were associated with this demand. The reviews should also assess producing countries readiness for the upcoming regulation on deforestation-forest products, based on which appropriate support measures could be designed together with producing countries' governments to help better prepare affected regions for complying with this regulation. Adequate and timely evidence will be required to make sure that the regulation does not have unintended impacts and to minimize them if such impacts are observed during the general reviews stipulated in the regulation.

4.5. Forest partnerships

Even though the text of the proposal does not specifically include any measures to address the potential disadvantages that smallholders and IPLCs could face after the regulation on deforestation-free products enters into force, funding for such measures could be linked to other EU development cooperation instruments. For instance, EU has pledged € 1 billion for Forest Partnerships (European Commission, 2021d), which was announced at the same time as the EU regulation proposal on deforestation-free products. Forest Partnerships will be a pillar of the European Green Deal and will aim "to facilitate protection, restoration and sustainable management of forests in the partner countries, for the benefit of people, climate and environment" (European Commission, 2021b, 2021d). According to the European Commission, Forest Partnerships will be a new development cooperation tool to help countries strengthen their forest governance and capacities to reduce deforestation (European Commission, 2021d). However, further concrete information on how and when this tool will start to function was not yet available during the time of writing this commentary.

5. Concluding remarks

The EU regulation proposal on deforestation-free products represents an ambitious plan to address deforestation and forest degradation linked to EU consumption of forest-risk commodities and their derivatives. The new rules are expected to bring a reduction in global greenhouse gas emissions and loss of biodiversity and reduce EU's impact on global deforestation and forest degradation linked to international trade. We recognize the important effort made to assemble the proposal. However, we argue that without adequate safeguards, the regulation can have unintended consequences for vulnerable groups in producing countries, such as smallholders, Indigenous people, and local communities. Such unintended impacts could include exclusion of smallholders and IPLCs from high-value supply chains of commodities covered by the regulation, insufficient price premiums or their unequal distribution to cover the costs of complying with this regulation, increase the risk of land conflicts between large-scale and small-scale agriculture, push smallholders and IPLCs to marginal lands, and even exacerbate human rights risks associated with large-scale commodity production. In order to avoid or minimize such impacts, we propose 1) to make provisions for companies, smallholders, and producing regions to receive sufficient support in their transition to deforestation-free supply chains and to provide incentives so that this transition can be inclusive, 2) to further promote and strengthen holistic approaches, such as jurisdictional or other integrated landscape management approaches to address the many issues that can arise from conflicting interests and unequal power distribution in affected landscapes; 3) to make clear provisions for protection of human rights and to promote land rights for smallholders and IPLCs in relation to tenure security; and 4) to ensure that regular and robust impact assessments inform policy about both direct and indirect impacts of the regulation on smallholders and IPLCs. It is essential that an effective and efficient incentive instrument is enforced in parallel to the regulation, e.g. within the framework of recently announced Forest Partnerships, as part of a participative process which involves smallholders and indigenous people at an early stage of planning. Timely and adequate action based on robust impact assessments can help minimize the unintended effects of this regulation and make sure that the progress achieved so far in terms of other UN SDGs related to poverty, inequality, hunger, and land restoration is not compromised.

Declaration of Competing Interest

We wish to confirm that there are no known conflicts of interest associated with this publication, titled "*Potential impacts of the proposed EU regulation on deforestation-free supply chains on smallholders, Indigenous peoples, and local communities in producer countries outside the EU*" and there has been no significant financial support for this work that could have influenced its outcome.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We confirm that we have given due consideration to the protection of intellectual property associated with this work and that there are no impediments to publication, including the timing of publication, with respect to intellectual property. In so doing we confirm that we have followed the regulations of our institutions concerning intellectual property.

We understand that the Corresponding Author is the sole contact for the Editorial process (including Editorial Manager and direct communications with the office). He/she is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs. We confirm that we have provided a current, correct email address which is accessible by the Corresponding Author and which has been configured to accept email (eliza.zhunusova@thuenen.de).

Acknowledgements

We are very grateful to two anonymous reviewers and the editorial team for their constructive comments on the previous version of this manuscript. The authors gratefully acknowledge the financial support from the Johann Heinrich von Thünen Institute, Federal Research Institute for Rural Areas, Forestry and Fisheries (THG fund) for Azin Sadeghi within the framework of the LaForeT-R^2 project and for Vianny Ahimbisibwe within the framework of the FLESRA project, as well as the financial support from the Alexander von Humboldt Foundation (Georg Forster fellowship) for Tarin Toledo-Aceves. Sen Le Thi Hoa would like to thank the Strong Research Group Program of Hue University for supporting this research. The views and opinions expressed in this article are those of the authors and do not necessarily reflect the official policy or position of their respective institutions.

References

Akinnifesi, F., Chirwa, P., Ajayi, O.O., Sileshi, G., Matakala, P., Kwesiga, F.R., Harawa, H., Makumba, W., 2008. Contributions of agroforestry research to livelihood of smallholder farmers in southern Africa: 1. Taking stock of the adaptation, adoption and impact of fertilizer tree options. Agric. J. 3, 58–75.

Alix-Garcia, J., Gibbs, H.K., 2017. Forest conservation effects of Brazil's zero deforestation cattle agreements undermined by leakage. Glob. Environ. Chang. 47, 201–217. https://doi.org/10.1016/j.gloenvcha.2017.08.009.

Anderson, T., 2006. Oil Palm and Small Farmers in Papua New Guinea. Report for the Centre for Environmental law and Community Rights on the Economic Prospects for Small Farmers in PNG's Oil Palm Industry. University of Sydney. https://np-net.pbworks.com/f/Anderson+(2006)+Oil+palm+and+small+farmers+in+PNG,+CELCR.pdf.

Ansah, E.O., Kaplowitz, M.D., Lupi, F., Kerr, J., 2020. Smallholder participation and procedural compliance with sustainable cocoa certification programs. Agroecol. Sustain. Food Syst. 44 (1), 54–87. https://doi.org/10.1080/21683565.2019.1579776.

Apriani, E., Kim, Y.-S., Fisher, L.A., Baral, H., 2020. Non-state certification of smallholders for sustainable palm oil in Sumatra, Indonesia. Land Use Policy 99, 105112. https://doi.org/10.1016/j.landusepol.2020.105112.

Bakhtary, H., Matson, E., Mikulcak, F., Streck, C., Thomson., 2020. Company Progress in Engaging Smallholders to Implement Zero-Deforestation Commitments in Cocoa and Palm Oil. Climate Focus, p. 61. https://www.climatefocus.com/publications/report-company-progress-engaging-smallholders-implement-zero-deforestation-commitments.

Balana, B., Oyeyemi, M.A., 2022. Agricultural credit constraints in smallholder farming in developing countries: evidence from Nigeria. World Develop. Sustain. 1, 100012 https://doi.org/10.1016/j.wds.2022.100012.

Balana, B., Mekonnen, D.K., Haile, B., Hagos, F., Yiman, S., Ringler, C., 2020. Are Smallholder Farmers Credit Constrained? Evidence on Demand and Supply Constraints of Credit in Ethiopia and Tanzania. International Food Policy Research Institute. https://doi.org/10.2499/p15738coll2.134152.

Benjaminsen, T.A., Holden, S., Lund, C., Sjaastad, E., 2009. Formalisation of land rights: some empirical evidence from Mali, Niger and South Africa. Land Use Policy 26 (1), 28–35. https://doi.org/10.1016/j.landusepol.2008.07.003.

Brack, D., 2019. Towards Sustainable Cocoa Supply Chains: Regulatory Options for the EU (P. 52). FERN, Tropenbos International, The Fairtrade Advocacy Office. https://www.fern.org/fileadmin/uploads/fern/Documents/2019/Fern-sustainable-cocoa-supply-chains-report.pdf.

Brandi, C., Cabani, T., Hosang, C., Schirmbeck, S., Westermann, L., Wiese, H., 2015. Sustainability standards for palm oil: challenges for smallholder certification under the RSPO. J. Environ. Dev. 24 (3), 292–314. https://doi.org/10.1177/1070496515593775.

Buchanan, J., Durbin, J., McLaughlin, D., McLaughlin, L., Thomason, K., Thomas, M., 2019. Exploring the reality of the jurisdictional approach as a tool to achieve sustainability commitments in palm oil and soy supply chains. Conserv. Intern. https://www.conservation.org/docs/default-source/publication-pdfs/jurisdictional_approach_full_report_march2019_published.pdf?Status=Master&sfvrsn=23c977ae_3.

Bulkan, J., 2020. Smallholder forestry in the FSC system: a review. Revue Gouvern. 17 (2), 7–29. https://doi.org/10.7202/1073109ar.

ClientEarth, 2021a. The Proposed EU Law on Deforestation-Free Products. What Is in the European Commission's Proposal and What Is Left out? ClientEarth, p. 17. https://www.clientearth.org/media/sbjhtw3c/eu-deforestation-proposal_main-elements-and-omissions_dec21.pdf.

ClientEarth, 2021b. Endorsing the End of the Amazon: Critical Weaknesses in the UK Government's Proposed forest Risk Commodities Framework and How to Fix Them. ClientEarth. https://www.clientearth.org/media/i1hmx1vc/endorsing-the-end-of-the-amazon_brazil-uk-policy-brief_10-09-21.pdf.

Coffee Board of India, 2020. Statistics on Coffee. Ministry of Commerce and Industry of India.. https://www.indiacoffee.org/coffee-statistics.html

Davis, B., Lipper, L., Winters, P., 2022. Do not transform food systems on the backs of the rural poor. Food Secur. https://doi.org/10.1007/s12571-021-01214-3.

De Sy, V., Herold, M., Achard, F., Avitabile, V., Baccini, A., Carter, S., Clevers, J.G.P.W., Lindquist, E., Pereira, M., Verchot, L., 2019. Tropical deforestation drivers and associated carbon emission factors derived from remote sensing data. Environ. Res. Lett. 14 (9), 094022 https://doi.org/10.1088/1748-9326/ab3dc6.

Dou, Y., da Silva, R., McCord, P., Zaehringer, J., Yang, H., Furumo, P., Zhang, J., Pizarro, J., Liu, J., 2020. Understanding how smallholders integrated into Pericoupled and Telecoupled systems. Sustainability 12 (4), 1596. https://doi.org/10.3390/su12041596.

Duguma, L.A., van Noordwijk, M., Bah, A., Duba, D., 2021. Tree commodities and climate change: impacts and opportunities. In: Minang, P.A., Duguma, L.A., van Noordwijk, M. (Eds.), Tree commodities and resilient green economies in Africa. World Agroforestry (ICRAF), Nairobi, Kenya.

Elahi, K., Michael, P., 2017. Oil Palm Plantation, Smallholders and Land Settlement Schemes in Papua New Guinea, 26, pp. 15–29.

Erbaugh, J.T., Pradhan, N., Adams, J., Oldekop, J.A., Agrawal, A., Brockington, D., Pritchard, R., Chhatre, A., 2020. Global forest restoration and the importance of prioritizing local communities. Nature Ecol. Evol. 4 (11), 1472–1476. https://doi.org/10.1038/s41559-020-01282-2.

Essen, M., Lambin, E.F., 2021. Jurisdictional approaches to sustainable resource use. Front. Ecol. Environ. 19 (3), 159–167. https://doi.org/10.1002/fee.2299.

European Commission, 2021a. Annexes to the Proposal for a Regulation on deforestation-free Products. Impact Assessment Report. https://ec.europa.eu/environment/publications/proposal-regulation-deforestation-free-products_en.

European Commission, 2021b. Forest Partnerships. https://ec.europa.eu/international-partnerships/topics/sustainable-forestry_en#header-690.

European Commission, 2021c. Proposal for a Regulation on Deforestation-free Products. EU. https://ec.europa.eu/environment/publications/proposal-regulation-deforestation-free-products_en.

European Commission, 2021d. Questions and Answers on New Rules for Deforestation-free Products. https://ec.europa.eu/commission/presscorner/api/files/document/print/en/qanda_21_5919/QANDA_21_5919_EN.pdf.

Fair Trade Advocacy Office, Fern, IUCN NL, Solidaridad, Tropenbos International, 2021. Including Smallholders in EU Action to Protect and Restore the World's Forests. Briefing Paper, p. 12. https://www.fern.org/publications-insight/including-smallholders-in-eu-action-to-protect-and-restore-the-worlds-forests-2387/.

FAO, 2020a. Global Forest Resources Assessment 2020. FAO, Key Findings. https://doi.org/10.4060/ca8753en.

FAO, 2020b. Global Forest Resources Assessment, 2020: Main Report. Food and Agriculture Organization of the United Nations. http://www.fao.org/3/ca9825en/ca9825en.pdf.

FERN, 2021. EU Regulation on Deforestation-free Products: What's in the New Proposal and What Does it Mean. FERN. https://www.fern.org/fileadmin/uploads/fern/Documents/2021/Fern_Deforestation-Regulation-briefing_01.pdf.

Garcia, M., Coletti, F., Banhe, A., Souza, G., Ouro, A., 2021. Jurisdictional Approaches. An Analysis of Brazil's States and Companies Contribution. CDP Latin America. https://cdn.cdp.net/cdp-production/cms/policy_briefings/documents/000/005/673/original/CDP_Brazil_JurisdictionalApproach_PolicyBrief_English.pdf?1617963574.

Garnett, S.T., Burgess, N.D., Fa, J.E., Fernández-Llamazares, Á., Molnár, Z., Robinson, C. J., Watson, J.E.M., Zander, K.K., Austin, B., Brondizio, E.S., Collier, N.F., Duncan, T., Ellis, E., Geyle, H., Jackson, M.V., Jonas, H., Malmer, P., McGowan, B., Sivongxay, A., Leiper, I., 2018. A spatial overview of the global importance of indigenous lands for conservation. Nature Sustain. 1 (7), 369–374. https://doi.org/10.1038/s41893-018-0100-6.

Gómez Tovar, L., Martin, L., Gómez Cruz, M.A., Mutersbaugh, T., 2005. Certified organic agriculture in Mexico: market connections and certification practices in large and small producers. J. Rural. Stud. 21 (4), 461–474. https://doi.org/10.1016/j.jrurstud.2005.10.002.

Grabs, J., Cammelli, F., Levy, S.A., Garrett, R.D., 2021. Designing effective and equitable zero-deforestation supply chain policies. Glob. Environ. Chang. 70, 102357 https://doi.org/10.1016/j.gloenvcha.2021.102357.

Guereña, A., Zepeda, R., 2013. The Power of Oil Palm: Land Grabbing and Impacts Associated with the Expansion of Oil Palm Crops in Guatemala: The Case of the Palmas del Ixcán Company, 2013. Oxfam America Research Backgrounder Series. http://www.oxfamamerica.org/publications/power-of-oil-palm-guatemala.

Herliana, S., Sutardi, A., Aina, Q., Himmatul Aliya, Q., Lawiyah, N., 2018. The constraints of agricultural credit and government policy strategy. MATEC Web of Conf. 215, 02008. https://doi.org/10.1051/matecconf/201821502008.

Hirsch, P., Mellac, M., Scurrah, N., 2015. The Political Economy of Land Governance in Viet Nam. Mekong Region Land Governance, p. 19 [Research Report]. https://halshs.archives-ouvertes.fr/halshs-02372282.

Hoang, H.T.N., Hoshino, S., Onitsuka, K., Maraseni, T., 2019. Cost analysis of FSC forest certification and opportunities to cover the costs a case study of Quang tri FSC group in Central Vietnam. J. For. Res. 24 (3), 137–142. https://doi.org/10.1080/13416979.2019.1610993.

Hosonuma, N., Herold, M., De Sy, V., De Fries, R.S., Brockhaus, M., Verchot, L., Angelsen, A., Romijn, E., 2012. An assessment of deforestation and forest degradation drivers in developing countries. Environ. Res. Lett. 7 (4), 044009 https://doi.org/10.1088/1748-9326/7/4/044009.

IDH, 2021. Coffee Production in the face of Climate Change. Country Profiles. Global Coffee Platform, p. 62. https://www.idhsustainabletrade.com/publication/coffee-production-in-the-face-of-climate-change/.

Ingram, V., Behagel, J., Mammadova, A., Verschuur, X., 2020. The Outcomes of Deforestation-free Commodity Value Chain Approaches. Forest and Nature Conservation Policy Group Wageningen, p. 88. URL. https://www.wur.nl/en/project/Outcomes-of-deforestation-free-commodity-value-chain-approaches.htm.

IPCC, 2019. In: Shukla, P.R., Skea, J., Buendia, E. Calvo, Masson-Delmotte, V., Pörtner, H.-O., Roberts, D.C., Zhai, P., Slade, R., van Connors, S., Diemen, R., Ferrat, M., Haughey, E., Luz, S., Neogi, S., Pathak, M., Petzold, J., Pereira, J. Portugal, Vyas, P., Huntley, E., Kissick, K., Belkacemi, M., Malley, J. (Eds.), Climate Change and Land: An IPCC Special Report on Climate Change, Desertification, Land Degradation, Sustainable Land Management, Food Security, and Greenhouse Gas Fluxes in Terrestrial Ecosystems: Summary for Policymakers. Intergovernmental Panel on Climate Change.

ITTO, FAO, 2020. Forest Governance and Climate-change Mitigation. A Policy Brief. https://forestindustries.eu/sites/default/files/userfiles/1file/fao-itto-climate%20change%20mitigation.pdf.

Jena, P.R., Stellmacher, T., Grote, U., 2017. Can coffee certification schemes increase incomes of smallholder farmers? Evidence from Jinotega, Nicaragua. Environ. Dev. Sustain. 19 (1), 45–66. https://doi.org/10.1007/s10668-015-9732-0.

Jong, H.N., 2020, July 7. Groups Demand Financial, Human Rights Probes into Palm Conglomerate Korindo. Mongabay Series: Indonesian Forests. Indonesian Palm Oil. https://news.mongabay.com/2020/07/financial-human-rights-probes-palm-oil-conglomerate-korindo-korea-papua-payment/.

Lowder, S.K., Skoet, J., Singh, S., 2014. What Do we Really Know about the Number and Distribution of Farms and Family Farms in the World? Background Paper for the State of Food and Agriculture 2014.

Lowder, S.K., Skoet, J., Raney, T., 2016. The number, size, and distribution of farms, smallholder farms, and family farms worldwide. World Dev. 87, 16–29.

Maertens, M., Minten, B., Swinnen, J., 2009. Growth in High-value Export Markets in Sub-Saharan Africa and Its Development Implications (LICOS Discussion Paper No. 245). Katholieke Universiteit Leuven, LICOS Centre for Institutions and Economic Performance. http://hdl.handle.net/10419/75037.

Marzano, K., 2021, December 15. Deforestation-free Commodity Chains: How an EU Legislative Proposal Reverberates in Brazil. Institute for Advanced Sustainability Studies. https://www.iass-potsdam.de/en/blog/2021/12/deforestation-free-commodity-chains-how-eu-legislative-proposal-reverberates-brazil.

Meyfroidt, P., Vu, T.P., Hoang, V.A., 2013. Trajectories of deforestation, coffee expansion and displacement of shifting cultivation in the central highlands of Vietnam. Glob. Environ. Chang. 23 (5), 1187–1198. https://doi.org/10.1016/j.gloenvcha.2013.04.005.

Neilson, J., 2008. Global private regulation and value-chain restructuring in Indonesian smallholder coffee systems. World Dev. 36 (9), 1607–1622. https://doi.org/10.1016/j.worlddev.2007.09.005.

Nguyen, V.Q., To, X.P., Nguyen, T.Q., Cao, T.C., 2018. Linking Smallholder Plantations to Global Markets: Lessons from the IKEA model in Vietnam. Forest Trends Report Series. Forest Policy, Trade, and Finance, p. 40. Forest Trends. https://www.forest-trends.org/wp-content/uploads/2018/06/IKEA-case-study-15-June_Final.pdf.

Okello, J.J., Narrod, C.A., Roy, D., 2011. Export standards, market institutions and smallholder farmer exclusion from fresh export vegetable high value chains: experiences from Ethiopia, Kenya and Zambia. J. Agric. Sci. 3 (4), 188. https://doi.org/10.5539/jas.v3n4p188.

Pacheco, P., 2012. Smallholders and communities in timber markets: conditions shaping diverse forms of engagement in tropical Latin America. Conserv. Soc. 10 (2), 114. https://doi.org/10.4103/0972-4923.97484.

Pacheco, P., Bakhtary, H., Camargo, M.C., Donofrio, S., Drigo, I., Mithöfer, D., 2018. The Private Sector: Can Zero Deforestation Commitments Save Tropical Forests? Transforming REDD+ Lessons and New Directions.

Pacheco, P., Mo, K., Dudley, N., Shapiro, A., Aguilar-Amuchastegui, N., Ling, P., Anderson, C., Marx, A., 2021. Deforestation Fronts: Drivers and Responses in a Changing World. WWF, Gland, Switzerland.

Pham, T.T., Nguyen, D.T., Đào Thi, L.C., Hoàng, T.L., 2020. Preparing Vietnam for new rules on international market: zero deforestation production and business. Center Intern. Forestry Res. (CIFOR). https://doi.org/10.17528/cifor/007573.

Ponte, S., 2002. The `latte revolution'? Regulation, markets and consumption in the global coffee chain. World Dev. 30 (7), 1099–1122. https://doi.org/10.1016/S0305-750X(02)00032-3.

Potter, L., 2020. Colombia's oil palm development in times of war and 'peace': myths, enablers and the disparate realities of land control. J. Rural. Stud. 78, 491–502. https://doi.org/10.1016/j.jrurstud.2019.10.035.

Proforest, 2016. Profiles of Company-Smallholder Working Arrangements in the Honduran Palm Oil Sector. https://www.proforest.net/resources/publications/profiles-of-palm-oil-company-smallholder-working-arrangements-13456/.

Reed, J., Deakin, L., Sunderland, T., 2015. What are 'integrated landscape approaches' and how effectively have they been implemented in the tropics: a systematic map protocol. Environ. Evid. 4 (1), 2. https://doi.org/10.1186/2047-2382-4-2.

Román-Dañobeytia, F., Levy-Tacher, S., Macario-Mendoza, P., Zúñiga-Morales, J., 2014. Redefining secondary forests in the Mexican Forest code: implications for management, restoration, and conservation. Forests 5 (5), 978–991. https://doi.org/10.3390/f5050978.

Sayer, J., Sunderland, T., Ghazoul, J., Pfund, J.-L., Sheil, D., Meijaard, E., Venter, M., Boedhihartono, A.K., Day, M., Garcia, C., van Oosten, C., Buck, L.E., 2013. Ten principles for a landscape approach to reconciling agriculture, conservation, and other competing land uses. Proc. Natl. Acad. Sci. 110 (21), 8349–8356. https://doi.org/10.1073/pnas.1210595110.

Schulte, I., Landholm, D.M., Bakhtary, H., Cabezas, S.C., Siantidis, S., Streck, C., 2020. Supporting Smallholder Farmers for a Sustainable Cocoa Sector: Exploring the Motivations and Role of Farmers in the Effective Implementation of Supply Chain Sustainability in Ghana and Côte d'Ivoire, p. 59. https://www.climatefocus.com/sites/default/files/Supporting%20Smallholder%20Farmers%20for%20a%20Sustainable%20Cocoa%20Sector%20June%202020.pdf.

Schurmann, A.T., Johnston, H.B., 2009. The group-lending model and social closure: microcredit, exclusion, and health in Bangladesh. J. Health Popul. Nutr. 27 (4), 518–527. https://doi.org/10.3329/jhpn.v27i4.3398.

Seymour, F.J., Aurora, L., Arif, J., 2020. The jurisdictional approach in Indonesia: incentives, actions, and facilitating connections. Front. Forests Global Change 3, 503326. https://doi.org/10.3389/ffgc.2020.503326.

SNV. (n.d.). REDD+ and Deforestation-free Agriculture in Vietnam. Retrieved February 28, 2022, from https://snv.org/assets/explore/download/redd_and_deforestation-free_agriculture_in_vietnam.pdf.

Tura, H.A., 2018. Land rights and land grabbing in Oromia, Ethiopia. Land Use Policy 70, 247–255. https://doi.org/10.1016/j.landusepol.2017.10.024.

United Nations, 1948. Universal Declaration of Human Rights. https://www.un.org/en/about-us/universal-declaration-of-human-rights.

VECO, 2011. Increased Incomes for Indonesian Cocoa Farmers in Sustainable Markets: NGO-Private Sector Cooperation on Sulawesi Island. Full Case Study (p. 21). Vredeseilanden Country Offices Indonesia. https://ei-ado.aciar.gov.au/sites/default/files/VECO(2011)IncreasedIncomesIndoCocoaFarmersSustainableMarketsSulawesi.pdf?_cf_chl_tk=3e8Cgcuw1fSv74jMG5H.6EfKHwfW9B0u8EC1wG.LL_s-1646672712-0-gaNycGzNCFE.

van der Ven, H., Rothacker, C., Cashore, B., 2018. Do eco-labels prevent deforestation? Lessons from non-state market driven governance in the soy, palm oil, and cocoa sectors. Glob. Environ. Chang. 52, 141–151. https://doi.org/10.1016/j.gloenvcha.2018.07.002.

Wessel, M., Quist-Wessel, P.M.F., 2015. Cocoa production in West Africa, a review and analysis of recent developments. NJAS - Wageningen J. Life Sci. 74–75, 1–7. https://doi.org/10.1016/j.njas.2015.09.001.

WWF, 2014. The Growth of Soy: Impacts and Solutions. WWF International, p. 96. https://awsassets.panda.org/downloads/wwf_soy_report_final_feb_4_2014_1.pdf.

WWF, 2021. Stepping Up? The Continuing Impact of EU Consumption on Nature Worldwide, p. 33. https://www.wwf.de/fileadmin/fm-wwf/Publikationen-PDF/WWF-Report-Stepping-up-The-continuing-impact-of-EU-consumption-on-nature-worldwide-FullReport.pdf.